

C02-AT2-Scenario-Based Questions

Course Name: COMPUTER VISION

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1. Low Contrast Image

A grayscale image appears dull with very little difference between object and background intensities.

- (a) Interpretation: The scenario describes an image lacking dynamic range. The cause of low contrast is typically poor lighting conditions, incorrect aperture settings during capture, or limitations of the imaging sensor, resulting in pixel intensities clustered in a narrow band.
- (b) Key Issues: The primary issues are poor visibility of details, difficulty in distinguishing objects from the background, and general degradation of both aesthetic quality and analytical usefulness.
- (c) Technique: Histogram Equalization.
- (d) Justification: Histogram Equalization is the best choice because it mathematically redistributes the pixel intensities across the full 0-255 range. This effectively stretches out the clustered values and enhances the global contrast without requiring any external data parameters.
- (e) Structure: Explanation provided logically above.

2. Uneven Illumination

An image captured under non-uniform lighting shows bright and dark patches.

- (a) Interpretation: The image has varying light levels across its area due to non-uniform lighting sources. This uneven illumination heavily impacts analysis because objects in shadows become too dark, while objects in glare become washed out.
- (b) Key Factors: Directional light sources, cast shadows, and curved surfaces reflecting light unevenly are the primary factors causing this intensity variation.
- (c) Technique: Contrast Limited Adaptive Histogram Equalization (CLAHE) or Homomorphic Filtering.
- (d) Justification: CLAHE divides the image into small tiles and applies equalization locally rather than globally, which perfectly normalizes illumination across different patches. Homomorphic filtering is also excellent as it separates and suppresses the low-frequency illumination component while preserving high-frequency details.
- (e) Structure: Explanation provided logically above.

3. Salt-and-Pepper Noise

A scanned document contains salt-and-pepper noise affecting readability.

- (a) Interpretation: The document is corrupted by random white (salt) and black (pepper) pixels. This is an impulse noise characterized by sudden, extreme variations in pixel values.
- (b) Key Issues: This noise directly obscures text characters, heavily reduces human readability, and causes severe errors in automated Optical Character Recognition (OCR) systems.
- (c) Technique: Median Filter.
- (d) Justification: A median filter replaces each noisy pixel with the median value of its neighboring pixels. It is highly suitable because it entirely eliminates extreme impulse values without blurring the sharp edges of the text, unlike standard mean/average filters.
- (e) Structure: Explanation provided logically above.

4. Blurred Edges

An image processed for noise removal appears overly smooth, causing edges to blur.

- (a) Interpretation: The image has lost its sharpness because the noise removal process (likely a linear low-pass filter) averaged out the high-frequency components that make up edges.
- (b) Key Issues: The loss of high-frequency spatial details makes boundary detection difficult, washes out structural features, and reduces overall image definition.
- (c) Technique: Unsharp Masking or High-Boost Filtering.
- (d) Justification: Unsharp masking mathematically subtracts a blurred version of the image from the original to isolate the edges (creating an edge mask). This mask is then added back to the original image, actively restoring and enhancing the lost edge sharpness.
- (e) Structure: Explanation provided logically above.

5. High-Frequency Noise

A satellite image contains fine-grained noise affecting feature visibility.

- (a) Interpretation: The satellite image contains rapid, random spatial changes in pixel intensities, representing high-frequency noise commonly introduced by transmission errors or sensor thermal noise.
- (b) Key Issues: This noise masks fine geographical features, degrades visual clarity, and disrupts critical feature extraction processes required for terrain mapping.
- (c) Technique: Butterworth Low-Pass Filter (Frequency Domain).
- (d) Justification: A Butterworth Low-Pass filter smoothly attenuates high-frequency noise without a sharp cutoff. This is vastly superior to an Ideal Low-Pass Filter because it avoids the

"ringing" effect (artifacts) that would further distort the satellite imagery.

(e) Structure: Explanation provided logically above.

6. Object Boundary Detection

In a medical image, boundaries between tissues need to be clearly identified.

(a) Interpretation: There is a critical need to isolate exact anatomical structures. Boundary detection is highly important here for isolating organs, identifying tumors, and ensuring accurate medical diagnosis.

(b) Key Challenges: Medical images typically suffer from low inherent contrast, high noise (like ultrasound speckle), and fuzzy/diffuse boundaries between different soft tissues.

(c) Technique: Canny Edge Detector.

(d) Justification: The Canny Edge Detector is robust against noise (due to its initial Gaussian smoothing) and uses double thresholding with hysteresis to link weak and strong edges. This guarantees continuous, single-pixel thin boundaries, which is essential for precise medical analysis.

(e) Structure: Explanation provided logically above.

7. Simple Segmentation

An image contains objects clearly distinguishable from the background based on intensity.

(a) Interpretation: The foreground objects and the background have entirely different brightness levels. The requirement is to binarize the image to completely separate the objects for further measurement or counting.

(b) Key Features: The image histogram will display distinct bimodal characteristics (two prominent peaks separated by a deep valley), indicating high contrast between the background and objects.

(c) Technique: Global Thresholding (Otsu's Method).

(d) Justification: Otsu's algorithm automatically determines the optimal threshold value by minimizing intra-class variance and maximizing inter-class variance, providing a perfect, automated separation without manual guesswork.

(e) Structure: Explanation provided logically above.

8. Complex Segmentation

An image contains multiple regions with overlapping intensity values.

(a) Interpretation: Simple thresholding fails here because different regions share similar pixel intensity values, making them indistinguishable on a purely global scale.

(b) Key Issues: Intensity overlap, complex textures, and potentially uneven lighting hide the true

boundaries of the objects, causing simple methods to merge separate objects.

(c) Technique: Region Growing or Watershed Algorithm.

(d) Justification: Region Growing starts from predefined seed points and groups adjacent pixels based on specific similarity constraints (not just raw global intensity). The Watershed algorithm can effectively separate overlapping objects by treating intensities as a topographic map, making both methods highly effective for complex overlaps.

(e) Structure: Explanation provided logically above.

9. Broken Edges

After applying edge detection, object boundaries appear discontinuous.

(a) Interpretation: The extracted boundaries have gaps. This happens when edge gradients are weak in certain areas, when noise disrupts the edge, or when the threshold in the edge detector was set too high.

(b) Key Issues: Discontinuous edges prevent the formation of closed contours, rendering subsequent shape analysis, object recognition, and area calculations impossible.

(c) Technique: Morphological Edge Linking (Dilation followed by Erosion/Closing) or Hough Transform.

(d) Justification: Morphological Closing operations expand the edge pixels to bridge small gaps and then shrink them back, effectively closing discontinuities and forming solid, continuous boundaries without altering the overall shape.

(e) Structure: Explanation provided logically above.

10. Combined Requirement

An industrial inspection system requires noise removal while preserving important edges for defect detection.

(a) Interpretation: The system faces a dual challenge: it must eliminate factory environment noise without smoothing out the fine, sharp edges of structural defects (like cracks or scratches).

(b) Key Issues: Traditional noise removal (like mean filtering) inherently blurs edges, while edge enhancement amplifies existing noise. Finding a balance is difficult.

(c) Technique: Bilateral Filtering followed by Canny Edge Detection.

(d) Justification: Bilateral filtering is specifically designed to average pixels based on both spatial closeness and intensity similarity. This means it aggressively smooths uniform noisy regions while strictly preserving sharp edges. Once cleaned, a Canny Edge Detector can perfectly isolate the structural defects.

(e) Structure: Explanation provided logically above.